

Binary words with the same characteristic polynomial: a block-reversal lemma and a census to length 22

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Abstract

For a binary word w let H_w be the tridiagonal matrix with w on the diagonal and ones beside it. Philip Weiss asked on MathOverflow when two words that are not reverses of each other give the same characteristic polynomial. We prove one lemma that covers every mechanism known so far: if a symmetric 2×2 matrix G is compatible with the transfer matrices of some blocks and with the context around them, the blocks can be put in reverse order without changing the polynomial. Reversal itself, the interleaving family from an earlier answer and a new ratio-class family are three cases. An exhaustive census to length 22 finds 822 classes of words with a common polynomial; all but 17 of them are explained by the lemma together with complementation, and 154 of the pairs need a matrix G outside both named families. We extend the count of distinct polynomials to $n = 22$. The lemma and its two families are checked in the Lean proof assistant.

1 The question

Write H_w for the $n \times n$ matrix with the letters $w_1, \dots, w_n \in \{0, 1\}$ on the diagonal, ones immediately above and below it, and zeros elsewhere, and $P_w(t) = \det(tI - H_w)$ for its characteristic polynomial. Reading the word backwards gives the same polynomial, since H_{w^*} is H_w with rows and columns in reverse order. Weiss [1] counted the distinct polynomials,

$$a_n = 2, 3, 6, 10, 20, 36, 72, 134, 270, 526, 1052, 2072, 4154, 8231, 16504, \dots,$$

observed that reversal is the only coincidence for $n \leq 7$, and asked for a formula for a_n and for a description of the other coincidences. The smallest ones occur at $n = 8$. Here is one of them:

$$P_{10110010} = P_{10010110}.$$

Stop and look for the reason before reading on. Both words start and end with 10 and contain 10 three times; what changes is the order of the two letters between the copies.

An earlier answer to the question [2] explains this example and a whole family like it. Our aim is a single statement that explains it together with every other coincidence we can find, and an honest list of the ones it does not explain.

2 Transfer matrices

Expanding $\det(tI - H_w)$ along its last row gives the three-term recurrence

$$P_{wcd} = (t - d) P_{wc} - P_w, \quad P_\emptyset = 1, \quad P_c = t - c,$$

for letters c, d , which is the recurrence of a continuant [3, §6.7]. It packages into 2×2 matrices. Put

$$T(z) = \begin{pmatrix} z & -1 \\ 1 & 0 \end{pmatrix}, \quad M_w = T(t - w_1) T(t - w_2) \cdots T(t - w_n), \quad e = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Then P_w is the top-left entry of M_w , that is $P_w = e^\top M_w e$, because the top row of $M_w T(t-c)$ is the top row of M_w pushed through the recurrence. Each $T(z)$ has determinant 1, so $\det M_w = 1$. We write the entries as

$$M_w = \begin{pmatrix} p & -\rho \\ q & -s \end{pmatrix};$$

here $p = P_w$, and q, ρ, s are the polynomials of w with its first letter, its last letter, or both removed (as noted in [2]; we use only the entries).

The point of the matrices is that concatenation becomes multiplication: $M_{AVB} = M_A M_V M_B$, so

$$P_{AVB} = x_A^\top M_V y_B, \quad x_A = M_A^\top e, \quad y_B = M_B e. \quad (1)$$

Read this as: the context A and B enters only through two vectors, and the middle V only through its matrix.

Reversal is already an instance of what follows. Let $D = \text{diag}(1, -1)$. Then $T(z)D = \begin{pmatrix} z & 1 \\ 1 & 0 \end{pmatrix}$ is symmetric, that is $T(z)D = DT(z)^\top$, and transposing $P_w = e^\top M_w e$ moves D through the product one letter at a time, which reverses the word.

3 The block-reversal lemma

Lemma 1. *Let G be a symmetric 2×2 matrix over a commutative ring, let $N_1, \dots, N_m$ be 2×2 matrices with $N_i G = G N_i^\top$ for every i , and let x, x', y, y' be vectors and λ, λ' scalars with*

$$Gx = \lambda y', \quad Gx' = \lambda' y.$$

Then $\lambda' x^\top N_1 \cdots N_m y = \lambda x'^\top N_m \cdots N_1 y'$.

Proof. Write $V = N_1 \cdots N_m$ and $V_{\text{rev}} = N_m \cdots N_1$. Moving G across one factor at a time gives $GV^\top = G N_m^\top \cdots N_1^\top = N_m G N_{m-1}^\top \cdots N_1^\top = \cdots = V_{\text{rev}} G$. Hence

$$\lambda' x^\top V y = x^\top V G x' = x'^\top (V G)^\top x = x'^\top G V^\top x = x'^\top V_{\text{rev}} G x = \lambda x'^\top V_{\text{rev}} y',$$

where the second step transposes a scalar and the third uses $G^\top = G$. □

The condition $NG = GN^\top$ says that NG is symmetric. For $N = M_Y$ with entries p', q', ρ', s' and $G = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$ it is one linear equation,

$$q'\alpha - (p' + s')\beta + \rho'\gamma = 0, \quad (2)$$

and each condition $Gx \parallel y'$ is linear in G as well. So, for given words, whether a suitable G exists is a question of linear algebra over $\mathbb{Q}(t)$.

Combining the lemma with (1), for words A, A', B, B' and blocks $Y_1, \dots, Y_m$:

Theorem 2. *If $G^\top = G$, $M_{Y_i} G$ is symmetric for every i , $Gx_A = \lambda y_{B'}$ and $Gx_{A'} = \lambda y_B$ with $\lambda \neq 0$ in $\mathbb{Z}[t]$, then $P_{AY_1 \cdots Y_m B} = P_{A' Y_m \cdots Y_1 B'}$.*

Figure 1 shows three examples, one for each of the families below.

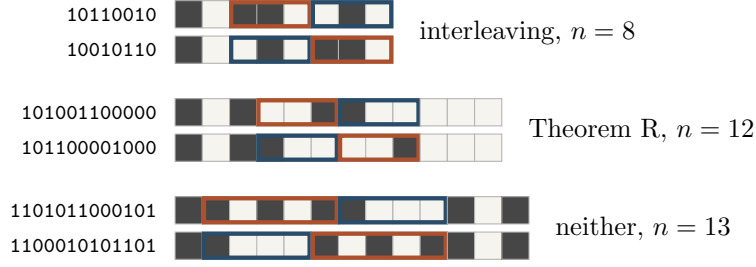


Figure 1: Three block reversals. Dark cells are 1, light cells 0; each block keeps its frame colour when the blocks are put in reverse order. In each pair the two words have the same characteristic polynomial and are not reverses of each other.

4 Three families

Interleaving. Fix a word W with entries p, q, ρ, s and letters $a_1, \dots, a_r$. The earlier answer [2] shows that $P_{Wa_1Wa_2\dots Wa_rW}$ depends on the letters only up to reversal.

Proposition 3. $P_{Wa_1W\dots a_rW} = P_{Wa_rW\dots a_1W}$. In Theorem 2 take $A = A' = W$, blocks a_iW , empty B ,

$$G = \begin{pmatrix} 1 - \rho^2 & -\rho p \\ -\rho p & -p^2 \end{pmatrix}, \quad \lambda = p.$$

Proof. With $x_A = (p, -\rho)^\top$ one finds $Gx_A = (p, 0)^\top = pe = py_B$. For a block, $M_{aW} = T(t - a)M_W$ has entries $p' = (t - a)p - q$, $q' = p$, $\rho' = (t - a)\rho - s$ and $s' = \rho$. Substituting into (2), the terms in $t - a$ cancel and the left side becomes $p(1 - (q\rho - ps)) = p(1 - \det M_W) = 0$. Finally $p = P_W$ is monic, so $\lambda \neq 0$. $\square$

At $n = 8$ this is the example from the introduction: $W = 10$ and the letters 1, 0 become 0, 1.

The ratio class. For a word Y with entries p', q', ρ', s' put $\kappa(Y) = (q' + \rho')/(p' + s')$.

Proposition 4 (Theorem R). *Let X be a word whose first letter c differs from its last letter, with entries p, q, ρ, s and $h = q + \rho$. Let A be X with its first letter flipped and B be X with its last letter flipped. If $\kappa(Y_i) = \kappa(X)$ for every i (for instance $Y_i \in \{X, X^*\}$), then $P_{AY_1\dots Y_m B} = P_{AY_m\dots Y_1 B}$.*

Proof. Take $G = \begin{pmatrix} p+s & h \\ h & p+s \end{pmatrix}$. By (2), $M_Y G$ is symmetric exactly when $(q' + \rho')(p + s) = (p' + s')h$, which is $\kappa(Y) = \kappa(X)$; in particular it holds for $Y = X$. Let $\varepsilon = 1$ if $c = 1$ and $\varepsilon = -1$ if $c = 0$, and $u = (1, \varepsilon)^\top$. Flipping the first letter changes $T(t - c)$ to $T(t - c) + \varepsilon ee^\top$, and since $e^\top = (0, 1)T(z)$ for every z , this gives $x_A = M_X^\top u$. In the same way, flipping the last letter gives $y_B = M_X u$. Now $Gu = (p + s + \varepsilon h)u$, so

$$Gx_A = GM_X^\top u = M_X Gu = (p + s + \varepsilon h)y_B,$$

using $M_X G = GM_X^\top$. Theorem 2 applies with $A' = A$, $B' = B$ and $\lambda = p + s + \varepsilon h$, which is monic of degree $|X|$ because p is and the other entries have smaller degree. $\square$

The first instances are at $n = 12$: with $X = 001$, $A = 101$, $B = 000$ and blocks 001, 100,

$$P_{101 001 100 000} = P_{101 100 001 000}.$$

The hypothesis on the end letters matters. If X starts and ends with the same letter, flipping the two ends gives the vectors $(1, \varepsilon)$ and $(1, -\varepsilon)$, which are different eigenvectors of G , and the argument fails; in random trials the identity then fails every time.

Complementation. Swapping 0 and 1 is not a reversal, but it relates polynomials in a simple way: $P_{\bar{w}}(t) = (-1)^n P_w(1 - t)$, as the earlier answer notes. The proof is one line: $T(t - \bar{c}) = -D T((1 - t) - c)D$ and $D^2 = I$. So $\bar{w}$ shares the polynomial of w whenever P_w is symmetric under $t \mapsto 1 - t$ (up to sign). The question's own example 00011011, 00100111 is of this kind.

Beyond the named families. Many coincidences need a G of neither shape. The smallest are at $n = 13$, for instance

$$P_{1\,10101\,1000\,101} = P_{1\,1000\,10101\,101},$$

with $A = A' = 1$, $B = B' = 101$, $\lambda = t^4 - 3t^3 + 4t - 1$ and $G = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$, where

$$\alpha = (t^3 - 2t^2 - t + 1)(t^3 - 2t^2 - t + 3), \quad \beta = (t - 1)(t^4 - 2t^3 - 2t^2 + 3t + 1), \quad \gamma = t(t - 2)(t^2 - 2).$$

The same G serves the blocks 10111 and 1000 in the same context, so the swap $1\,10111\,1000\,101 \sim 1\,1000\,10111\,101$ holds as well.

5 The census

We enumerated all words of length $n \leq 22$, grouped them by polynomial (fingerprinted at two points modulo $2^{61} - 1$, then every group confirmed with exact integer polynomials), and counted the classes that contain more than one reversal pair. For each class we joined two of its reversal pairs when one is the complement of the other, or when some splitting of the two words and some G satisfy Theorem 2. The conditions on G are linear, so the search solves them modulo a large prime at random values of t ; every move it found was then confirmed exactly over $\mathbb{Q}(t)$. A class counts as explained when its reversal pairs are all joined.

n	a_n	excess $r_n - a_n$	classes	unexplained
8	134	2	2	0
9	270	2	2	2
10	526	2	2	0
11	1 052	4	4	0
12	2 072	8	8	0
13	4 154	6	6	2
14	8 231	25	25	0
15	16 504	8	8	0
16	32 856	40	40	0
17	65 764	28	28	2
18	131 249	79	78	1
19	262 604	52	52	4
20	524 606	194	194	2
21	1 049 560	40	40	0
22	2 097 843	333	333	4

Table 1: Distinct polynomials a_n , the excess over the number $r_n = (2^n + 2^{\lceil n/2 \rceil})/2$ of words up to reversal, the classes of reversal pairs with a common polynomial, and the classes not explained by complementation and block reversal. Every class holds two reversal pairs except one at $n = 18$, which holds three.

Table 1 gives the result. The values of a_n for $n \leq 15$ agree with Weiss's, and those for $16 \leq n \leq 22$ are new; the sequence is not in the OEIS (searched 10 October 2026). Of the 822 classes, 805 are explained. Counting pairs of reversal pairs joined by a block reversal, 123 are interleavings, 56 have a G with equal diagonal entries as in Theorem R, and 154 need some other G .

The excess stays small: $(r_n - a_n)/2^{n/2}$ lies between 0.02 and 0.2 for $8 \leq n \leq 22$, which suggests $a_n = 2^{n-1} + O(2^{n/2})$, but we have no proof.

6 What is left

Seventeen classes are not explained. The first is

$$P_{111010100} = P_{100110110} \quad (n = 9),$$

together with its complement $110101000 \sim 100100110$. Its polynomial is irreducible of degree 9, and no splitting of the two words into context and blocks admits a G as in Theorem 2. The others are at $n = 13, 17, 18$ (the class with three reversal pairs), 19, 20 and 22. So the answer to Weiss’s second question is, as far as length 22: complementation and block reversal, plus a sporadic remainder that neither explains. Whether the remainder comes from a further mechanism, or the block-reversal lemma has a natural generalisation that reaches it, is open, as is a formula for a_n .

Verification. Lemma 1, Theorem 2, Propositions 3 and 4, the complement identity, the recurrence for P and the degree facts used to cancel λ are proved in Lean 4 with Mathlib [4], using only the standard axioms; there P_w is defined as the top-left entry of M_w , and its agreement with the determinant is the classical expansion quoted above. The census is a computation: the enumeration, the exact confirmation of every class, the search for G and the exact confirmation of all 333 moves it found are in Python and C, together with an exhaustive check for $n \leq 8$ that the search never accepts a move between words with different polynomials, deliberately broken versions of each program, and an independent program that recounts a_n for $n \leq 18$ and redoes the census for $n \leq 16$ in exact rational arithmetic. Everything is at <https://github.com/jvvk/mathematics/tree/main/binary-word-polynomials>.

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References

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